

Structural Centrality of British Rulers in the European Royal Kinship Network, 1495-1918

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Abstract

We study the structural centrality of British rulers within the evolving European royal kinship network between 1495 and 1918. This approach allows us to quantitatively assess the structural importance of British rulers. In addition, it allows us to measure and compare the centrality of British rulers over time. Britain's position within the European dynastic network was not consistently central, but instead evolved with changing patterns of intermarriage and political relationships over time. British rulers' centrality increased during the 16th and 17th centuries. They peaked in the 17th century. Then, they declined in the 18th century. By the 19th century, centrality levels for British and non-British rulers converged, showing a more balanced and interconnected European dynastic network.

Keywords: Historical network analysis, Network centrality, Social network analysis

1 Introduction

The British Empire was the leading global power in the 19th and early 20th centuries and is the largest empire in history [Ferguson \(2004\)](#). Its expansion was driven not only by military and economic strength, but also by diplomatic relationships and dynastic connections across Europe. European royal families were interconnected through marriage, which created alliances and shaped political influence. Similarly, other elite families such as Medici [Padgett and Ansell \(1993\)](#) leveraged how interconnecting with other families and strategic positioning yourself could influence political power. Previous research [Benzell and Cooke \(2021\)](#) has focused on whether family ties between

rulers reduce conflict between countries; little is known about how structural centrality metrics of British rulers politically influenced other European nations in allowing the formation of a British Empire.

We address this gap by asking: How did the structural centrality of British rulers within the European royal kinship network evolve between 1495 and 1918? To answer this question, we use genealogical and political data on European rulers of Christian Europe compiled by [Benzell and Cooke \(2021\)](#), which link royal kinship networks to sovereign monarchies over time.¹

2 Related Work

This article contributes to a large interdisciplinary literature on historical network analysis, network centrality, and social network analysis. The concept of centrality originated in [Freeman \(1978\)](#). Formally, the centrality metrics that we use are found in [Opsahl et al. \(2010\)](#) (weighted degree), [Bonacich \(1972\)](#) (eigenvector), [Freeman \(1977\)](#) (betweenness), and [Freeman \(1978\)](#) (closeness).

3 Methods

We construct a ruler-level network in which nodes represent European rulers whose reigns overlap the period under study, and edges represent genealogical relationships between rulers, derived from parent-child and marriage ties. The network is constructed as a projection of the full kinship graph. Adjacent edges between rulers are derived from the underlying kinship network of all individuals, such that rulers may be indirectly connected through non-ruling family members. The edges between rulers are weighted by the inverse kinship distance ($\frac{1}{d}$), where the distance is defined in [Benzell and Cooke \(2021\)](#) as the shortest path between individuals in the kinship network. The inverse distance $\frac{1}{d}$ takes values between zero and one. A disconnected kinship is defined as $\frac{1}{\infty}$, zero weight. Shorter genealogical distances correspond to stronger dynastic connections.

To evaluate structural position, we use network centrality metrics such as weighted degree, eigenvector, betweenness, and closeness centrality. We use these metrics because we believe they capture best the different aspects of the importance of a ruler within an evolving network over time in relation to the strength of connections they have, their role as intermediaries, and their overall integration within the broader dynastic system.

The network was constructed and the network centrality metrics were implemented using the Python library called NetworkX.

4 Results and Discussion

Table 1 reports the centrality metrics for British and non-British rulers over four historical periods. Across the multiple metrics, British rulers show high centrality in

¹Based on geography, this corresponds to about continental Europe, the British Isles, the Mediterranean Islands, and Russia.

the earlier periods, especially in the 16th and 17th centuries. For example, eigenvector centrality is consistently higher for British rulers than for non-British rulers in these periods, peaking in 1600–1699. Also, British rulers in the 17th century played a strong role as intermediaries within the dynastic network, given their betweenness centrality mean.

Table 1 Centrality Metrics of British vs Non-British Rulers Over Time

Period	Metric	British Mean	Non-British Mean
1495–1599	Weighted Degree	18.23	19.40
1495–1599	Eigenvector	0.119	0.108
1495–1599	Betweenness	0.014	0.018
1495–1599	Closeness	0.091	0.094
1600–1699	Weighted Degree	21.15	20.33
1600–1699	Eigenvector	0.144	0.121
1600–1699	Betweenness	0.027	0.015
1600–1699	Closeness	0.098	0.093
1700–1799	Weighted Degree	15.87	19.92
1700–1799	Eigenvector	0.092	0.118
1700–1799	Betweenness	0.010	0.017
1700–1799	Closeness	0.089	0.095
1800–1918	Weighted Degree	23.70	22.10
1800–1918	Eigenvector	0.090	0.084
1800–1918	Betweenness	0.0068	0.0066
1800–1918	Closeness	0.095	0.094

However, this pattern changed in the 18th century. The British centrality declines across all metrics. In particular, both eigenvector and betweenness centrality fall below those of non-British rulers, suggesting a weakening of Britain’s structural position within the network. In the 19th century and onward, centrality measures for British and non-British rulers converge. British rulers still hold a slightly higher weighted degree and eigenvector centrality. Nevertheless, the differences between the British and non-British rulers are small, we see a more balanced and interconnected European dynastic network.

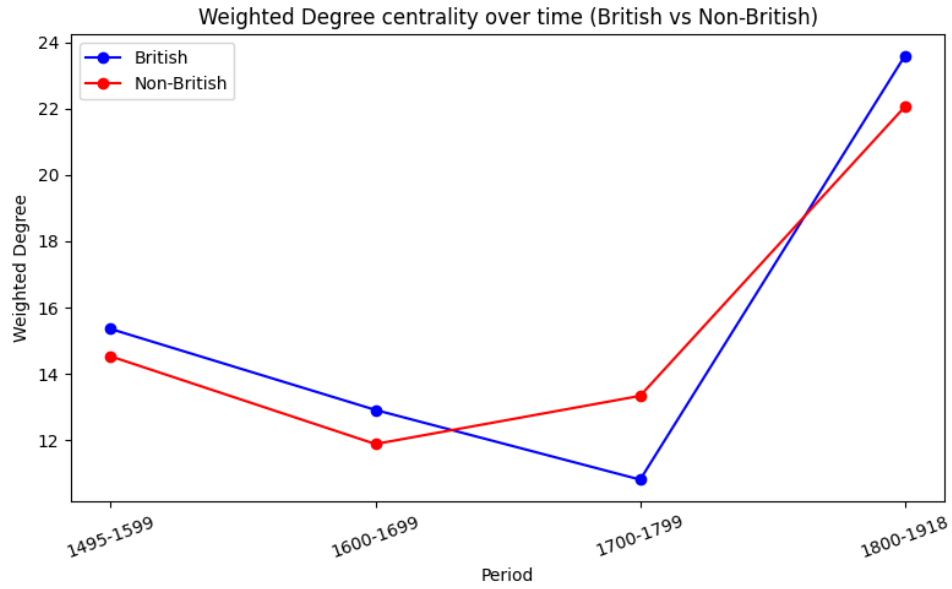


Fig. 1 Evolution of weighted degree centrality for British and non-British rulers across four historical periods (1495–1918).

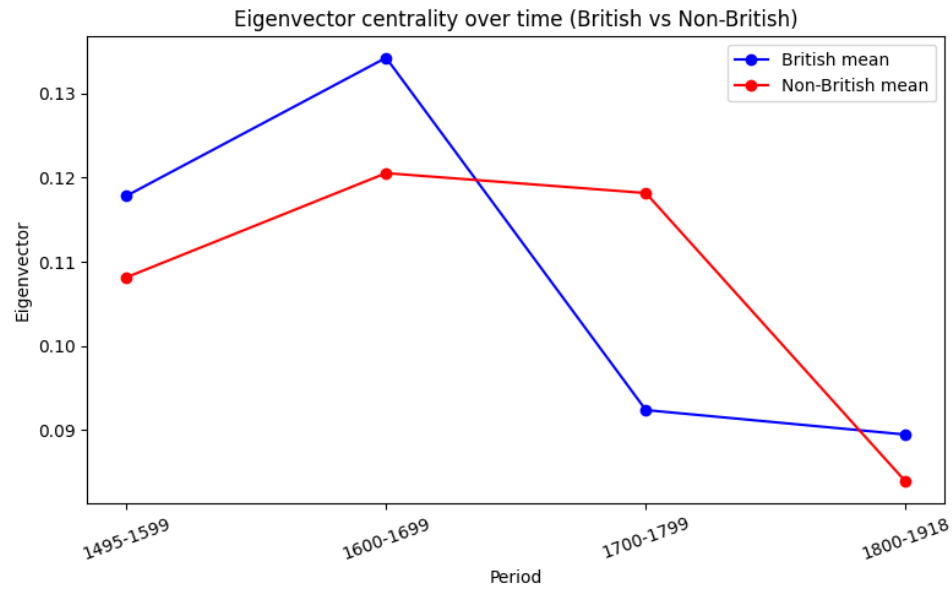


Fig. 2 Evolution of eigenvector centrality for British and non-British rulers across four historical periods (1495–1918).

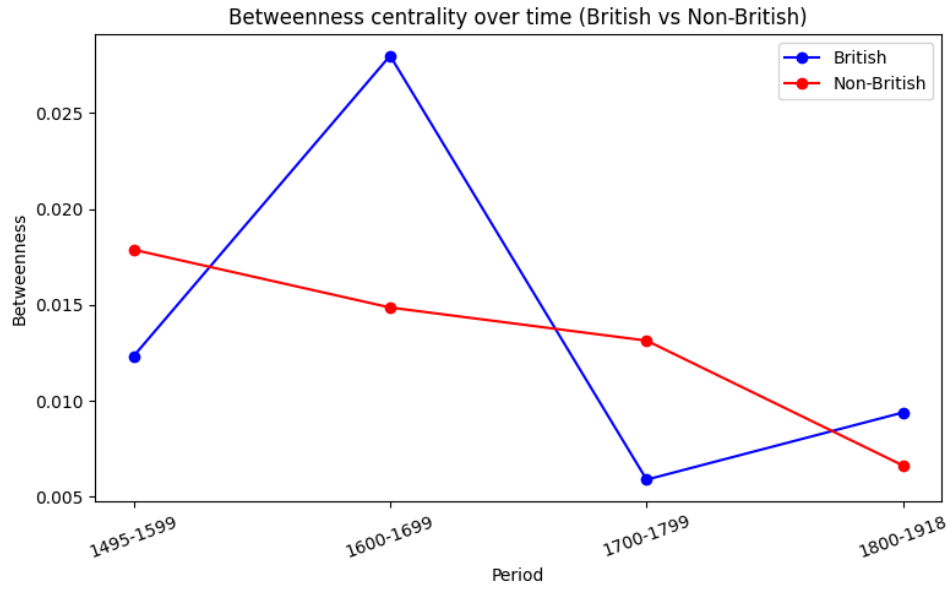


Fig. 3 Evolution of betweenness centrality for British and non-British rulers across four historical periods (1495–1918).

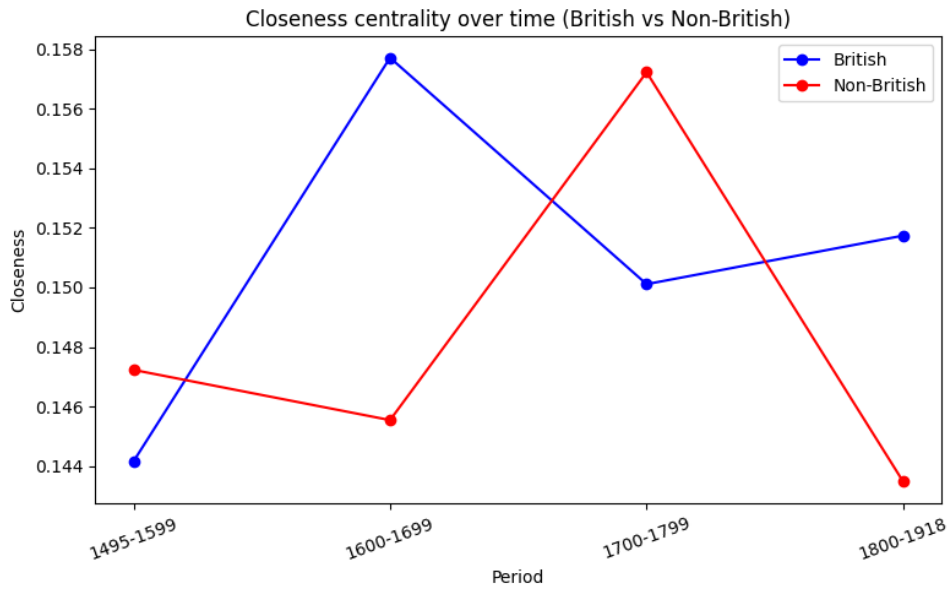


Fig. 4 Evolution of closeness centrality for British and non-British rulers across four historical periods (1495–1918).

As shown in Figures 1 and 4, we see relatively high score for the British in the metrics of weighted degree and closeness due to Queen Victoria’s grandchildren being superpowers in the 20th century, i.e. King George V of the United Kingdom, Czar Nicholas II of Russia, and Kaiser Wilhelm II of Germany. In Figure 2, the evolution of eigenvector centrality reinforces the patterns observed in Table 1. In Figure 3, we see that the British played an important role as an intermediary in the 17th century. The reason being likely due to the Union of the Crowns in 1603, which linked the English and Scottish dynastic networks under a single monarch, James VI and I, hence increased Britain’s connectivity within the broader European system [Ferguson \(1994\)](#).

British rulers show to have higher centrality in the 16th and 17th centuries, then a decline starting in the 18th century and beyond. In the 19th century, centrality levels converged between British and non-British rulers.

Overall, the results suggest that British rulers were not consistently central as a whole throughout many centuries. Their structural peak was the highest in the 17th century. As time went on, there was a decline in the influence British rulers had and there was a convergence with other European dynasties.

5 Future Work

Future work includes extending the analysis beyond Europe or beyond 1918, which could help assess whether similar patterns exist at other times in history. In addition, alternative metrics or models could be used or developed to better understand the importance the British rulers had in shaping to become a leading global superpower.

6 Conclusion

This study examined how the structural centrality of British rulers within the European royal kinship network evolved between 1495 and 1918. we find that British rulers were not consistently central across history. British rulers’ centrality increased during the 16th and 17th centuries. They peaked in the 17th century. Then, they declined in the 18th century. By the 19th century, centrality levels for British and non-British rulers converged, showing a more balanced and interconnected European dynastic network. These findings show that Britain’s position within the European dynastic network was not consistently central, but instead evolved over time with changing patterns of intermarriage and political relationships.

7 Availability of data and materials

All data applicable to this article are on GitHub.²

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²https://github.com/evanalba/british_structural_centrality_1495-1918

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